VIKUM BANDARA

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Email: vbandar@clemson.edu, vikumb@sci.pdn.ac.lk **Phone**: +1 (864) 765-4259
Location: Central, SC 29630, USA **Alternate Phone**: +94 77 270 6342
Website: Personal website

Summary

Mathematics researcher and educator with graduate-level teaching experience and research in combinatorics, discrete dynamical systems, algebraic/combinatorial structures, cryptography, and graph theory. Published work on Collatz-driven cryptography and symmetric encryption using snake graphs and supermagic covering. Experienced in using Python for computational experiments, visualization, and conjecture testing.

Education and Training

Clemson University Clemson, SC, USA
Master of Science in Mathematical Sciences (Combinatorics focus) December 2025
GPA: 3.57/4.00. 18 graduate-level courses.
Doctoral qualification: Passed the Optimization Research preliminary examination for PhD candidacy.
Relevant coursework: abstract algebra, combinatorics, graph theory, dynamical systems, optimization, cryptography, and scientific computing.

University of Peradeniya Peradeniya, Sri Lanka
Bachelor of Science (Hons) in Mathematics September 2022
Major in Mathematics (Pure and Applied), Minor in Physics.
Math GPA: 3.77/4.0, Overall GPA: 3.66/4.0. 4-year degree.

G.C.E. Advanced Level (Physical Science) Sri Lanka
Mathematics A, Physics B, Chemistry C December 2016

IELTS Academic in 2022
Overall 7.0 (Reading 8.0, Writing 6.5, Listening 7.0, Speaking 6.5).

Research Interests

Algebraic and combinatorial structures in discrete dynamical systems; orbit structure and permutation actions; labeled configurations; enumerative combinatorics; graph theory; cryptography; number theory; computational conjecture testing.

Publications

R. Ranasinghe and **V. Bandara**, "Collatz conjecture–driven cryptographic framework," *Journal of Discrete Mathematical Sciences & Cryptography*, vol. 29, no. 7, pp. 2621–2640, 2026. doi:10.47974/JDMSC-2232.

P. G. R. S. Ranasinghe, **R. M. V. V. Bandara**, and A. M. S. L. K. Athapaththu, "Symmetric encryption using snake graphs and supermagic covering," *Journal of the National Science Foundation of Sri Lanka*, vol. 52, no. 4, pp. 435–440, 2024. doi:10.4038/jnsfsr.v52i4.12196.

V. Bandara and S. Poznanović, "Bulgarian Solitaire with distinguishable cells," manuscript in preparation.

V. Bandara and R. Ranasinghe, "Novel Cryptographic Scheme Based on the Collatz Conjecture," *RESCON Research Conference*, 2022.

Research

Research Project in Combinatorics, Clemson University Clemson, SC
Read the complete master's project (PDF).

- Investigated generalizations of Bulgarian solitaire with Dr. Svetlana Poznanović.
- Used Python to visualize orbits and identified the phenomenon that a configuration can return to its original shape before individual cards return to their original positions.
- Developed a coloring algorithm on Ferrers diagrams connecting labeled-cell movement to permutation cycles and orbit sizes.
- Preparing manuscript *Bulgarian Solitaire with distinguishable cells* for publication.

Final-Year Research Project, University of Peradeniya Peradeniya, Sri Lanka

- Studied relationships among Collatz sequences with Dr. Rajitha Ranasinghe.
- Implemented a proposed cryptographic scheme based on Collatz sequence behavior using Python.
- Resulted in the publication "Collatz conjecture–driven cryptographic framework."

Undergraduate Research Projects, University of Peradeniya Peradeniya, Sri Lanka

- **Cryptographic Scheme Based on Graph Theory** (Oct 2022 – 2023): Studied supermagic labeling on wheel graphs with snake vertex labeling for cryptographic applications. Resulted in the publication "Symmetric encryption using snake graphs and supermagic covering" with P. G. R. S. Ranasinghe and A. M. S. L. K. Athapaththu (2024).
- **Constructive Approach on the Collatz Conjecture** (Jun 2021 – 2023): Constructed a novel structure consisting of Collatz sequences; investigated why Collatz sequences do not become periodic; showed the constructed structure contains all odd positive integers.
- **Novel Cryptographic Scheme Based on the Collatz Conjecture** (Jun 2021 – 2023): Developed a chaos-based symmetric encryption/decryption algorithm; resulted in publication at RESCON 2022.

Experience

Graduate Teacher of Record, Clemson University — Clemson, SC
January 2025 – Current
MATH 1020 – Business Calculus

Graduate Teaching Assistant, Clemson University — Clemson, SC
August 2023 – December 2024
MATH 1040 – Precalculus, MATH 1080 – Calculus, MATH 2080 – PDE, MATH 3110 – Linear Algebra

Teaching Assistant, University of Peradeniya — Peradeniya, Sri Lanka
September 2022 – April 2023

Venue Coordinator/Presenter, SURS Science Undergraduate Research Symposium — 2022

Academic Coordinator/Presenter, RESCON Research Conference — 2022

Skills

Programming
Proficient in: Python (computational experiments, visualization, cryptographic implementation), LaTeX.
Familiar with: RStudio, MATLAB, Julia, Gurobi.

Languages
English, Sinhala.

References

Dr. P. G. R. S. Ranasinghe, Head of the Department and Senior Lecturer, Department of Mathematics, University of Peradeniya, Sri Lanka.
Contact: +94 70 558 0514, rajithamath@sci.pdn.ac.lk.

Dr. Svetlana Poznanović, Professor and Associate Director for Mathematics, School of Mathematical and Statistical Sciences, Clemson University, SC 29634, USA.
Contact: +1 864 656 0412, spoznan@clemson.edu